SE LAB – 07

Name: Pavithra Reddy T
SRN: PES1UG24CS489
SEC: 5TH sem, I sec

Lab 7 – Fuzz Testing

Objective: Use Hypothesis to generate unexpected inputs for the three functions in processor.py, identify cases that cause exceptions, add input handling, and run the tests again.

Functions tested

sanitize_meter_id()	parse_consumption_readings()	format_billing_summary()

1. Original application

The original program was run with valid sample inputs.

```
C:\Users\lenovo\OneDrive\□ □ □ □ □ □ \se_labs\LAB06\L6_L7_01_water_utility_billing\Lab 7>python processor.py
==== Running sanitize_meter_id ====
Enter a meter ID with prefixes (!,@,#,$,%): !@#MTR-102
Sanitized Meter ID: MTR-102

==== Running parse_consumption_readings ====
Enter consumption readings (e.g. 10,25,30): 10,25,30
Parsed Readings: [10, 25, 30]

==== Running format_billing_summary ====
Enter account record (Name:Usage:Zone): John:100:North
Billing Summary: Customer John billed 200 USD in North

C:\Users\lenovo\OneDrive\□ □ □ □ □ □ \se_labs\LAB06\L6_L7_01_water_utility_billing\Lab 7>_
```

2. Fuzz tests before fixing

All three tests failed. Hypothesis found inputs that exposed missing input validation.

Function / input	Exception	Cause
sanitize_meter_id(None)	AttributeError	None has no strip()
parse_consumption_readings('')	ValueError	Empty value cannot convert to int
format_billing_summary('')	IndexError	Required fields are missing

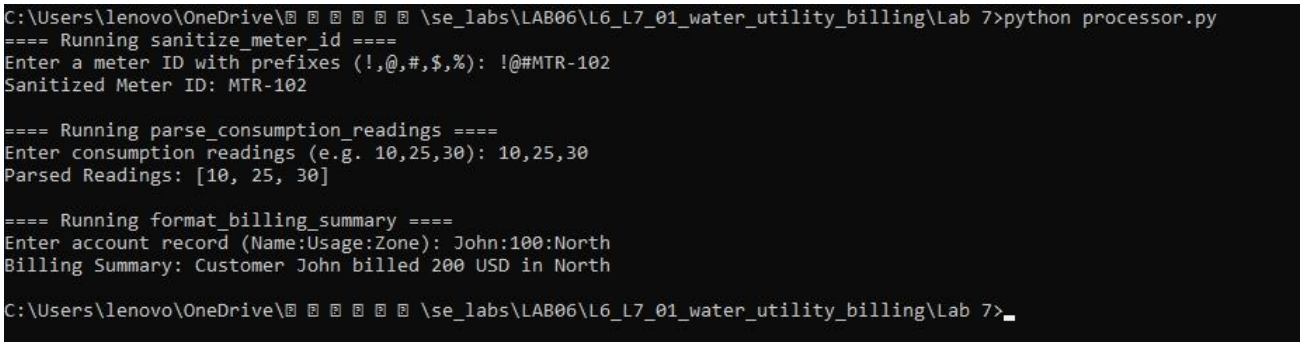
Initial test result: 3 failed

After the changes

Input checks and exception handling were added to the three functions. The same valid inputs still work, while the invalid examples above are handled without uncaught exceptions.

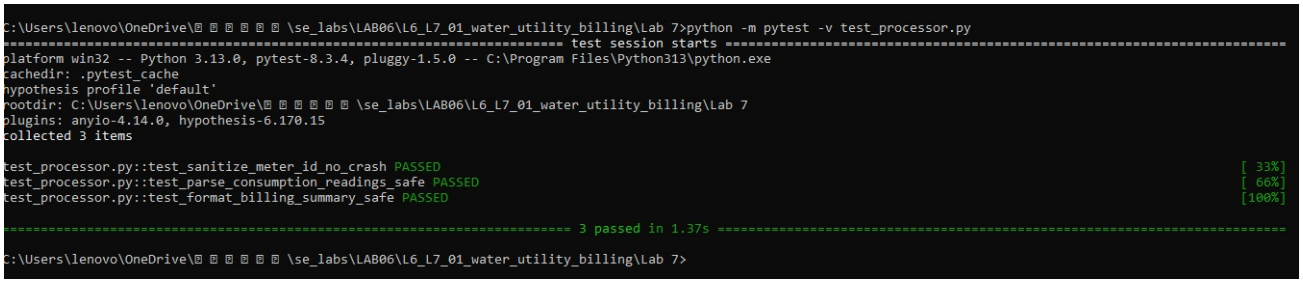
3. Fixed application

The program was run again using the same valid sample inputs.



4. Hypothesis tests after fixing

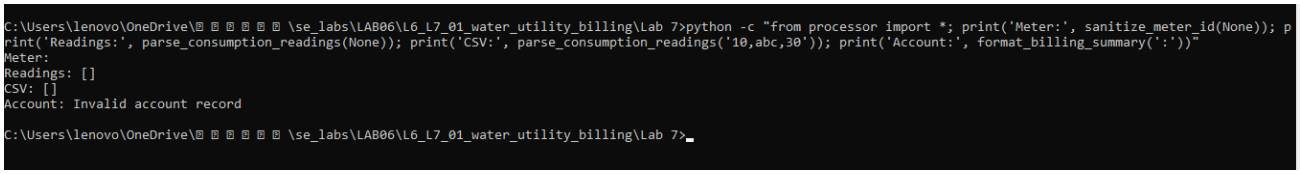
The same three tests were rerun against the updated processor.py.



Final test result: 3 passed

5. Additional invalid-input check

The fixed functions were called directly with invalid values to confirm that they return safe results.



Files: processor.py (fixed application) and test_processor.py (Hypothesis tests).